

**Lab 1: Familiarization with Signal Operations**

**Instructions:**

1. Lab report to be submitted (on Nalanda) at the end of the lab hours (preferably), late submission will result in loss of marks, use the lab hours to finish the work
  2. All the tasks in each lab have to be in one main MATLAB file with “.m” extension, differentiate each question by commenting. Alternatively, you can zip all .m files together, name them appropriately and upload one zip/rar file. Make one Word/pdf file with all the results corresponding to the questions
  3. Plots should include x label, y label, legend, and titles
  4. File name: ID\_Lab1.m/ID\_Lab1.rar or zip **and** ID\_Lab1.pdf
  5. If the report/code is found to be copied, zero marks will be awarded **[Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.]**
- 

1. Let  $x(n) = (0. p)^n u(n)$ ,  $h(n) = (-0. q)^n u(n)$ , and  $y(n) = h(n) * x(n)$ , where  $*$  denotes convolution operation,  $p$  and  $q$  are the second last and the last digit of your ID, respectively, if any of the digit is zero add 6 to that digit. Use 3 columns and 1 row of subplots for the following parts.
  - a) Truncate  $x(n)$  and  $h(n)$  to 28 samples. Use the **conv** function to compute  $y(n)$ . Plot  $y(n)$  using the **stem** function.
  - b) Using the **filter** function, determine the first 51 samples of  $x(n) * h(n)$ . Plot  $y(n)$  using the **stem** function. Compare your results with those of part a). What are your observations?
2. When the sequences  $x(n)$  and  $h(n)$  are of finite duration  $N_x$  and  $N_h$ , respectively, their linear convolution can be implemented using matrix-vector multiplication. If elements of  $y(n)$  and  $x(n)$  are arranged in column vectors  $\mathbf{x}$  and  $\mathbf{y}$ , respectively, then linear convolution can be obtained as  $\mathbf{y} = \mathbf{H}\mathbf{x}$ , where linear shifts in  $h(n-k)$  for  $n = 0, \dots, N_h - 1$  are arranged as rows in the matrix  $\mathbf{H}$ . This matrix has an interesting structure and is called a **Toeplitz matrix**.

To investigate this matrix, consider the following sequences

$$x(n) = \{1, 2, 3, 4, 5\} \text{ and } h(n) = \{p, q, r, s\}$$

↑

↑

where  $p, q, r$ , and  $s$  are the last 4 digits of your ID; if any digit is zero, add 8 to that digit.

- a) Express  $x(n)$  as a  $5 \times 1$  column vector  $\mathbf{x}$  and  $y(n)$  as an  $8 \times 1$  column vector  $\mathbf{y}$ . Now, determine the  $8 \times 5$  matrix  $\mathbf{H}$  so that  $\mathbf{y} = \mathbf{H}\mathbf{x}$ .
  - b) Characterize the matrix  $\mathbf{H}$ . From this characterization, can you define a Toeplitz matrix? How does this definition compare with that of time invariance?
  - c) What can you say about the first column and the first row of  $\mathbf{H}$ ?
-